

Real-world (build and compare the supervisor version) — Solution

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All three depths below reuse `research()`, `write()`, and `StageResult` from `sequential_measure`'s solution unchanged. Read all three depths — they're not "wrong, less wrong, right," they're 3 real, valid ways to solve the same problem, with real tradeoffs between them.

Basic Version

Approach 1 — the direct way, `if`-based routing

```
from typing import TypedDict
from langgraph.graph import StateGraph, START, END
from langgraph.types import Command

class SupervisorState(TypedDict):
    task: str
    research_text: str
    summary: str
    routing_log: list
    total_tokens: int
    total_seconds: float

def supervisor(state):
    if not state.get("research_text"):
        return Command(update={"routing_log": state["routing_log"] + ["research"]})
    if not state.get("summary"):
        return Command(update={"routing_log": state["routing_log"] + ["write"]},
                        goto=END)

def research_node(state):
    result = research(state["task"])
    return Command(
        update={
```

```

        "research_text": result.text,
        "total_tokens": state["total_tokens"] + result.tokens,
        "total_seconds": state["total_seconds"] + result.seconds,
    },
    goto="supervisor",
)

def write_node(state):
    result = write(state["research_text"])
    return Command(
        update={
            "summary": result.text,
            "total_tokens": state["total_tokens"] + result.tokens,
            "total_seconds": state["total_seconds"] + result.seconds,
        },
        goto="supervisor",
    )

builder = StateGraph(SupervisorState)
builder.add_node("supervisor", supervisor)
builder.add_node("research_node", research_node)
builder.add_node("write_node", write_node)
builder.add_edge(START, "supervisor")
graph = builder.compile()

result = graph.invoke({
    "task": "climate change",
    "research_text": "",
    "summary": "",
    "routing_log": [],
    "total_tokens": 0,
    "total_seconds": 0.0,
})
print(result["routing_log"])
print(result["total_tokens"], "tokens,", result["total_seconds"], "seconds")

```

Expected output (exact numbers vary by run):

```

['research', 'write']
306 tokens, 2.81 seconds

```

This works and correctly reuses `sequential_measure`'s functions unchanged. It's missing a fair, run-both comparison against the sequential pipeline, and its routing decision costs nothing — both fine for a first working version.

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Intermediate Version

Approach 1 — a `run_supervisor()` report matching `run_sequential()`'s shape

```
def run_supervisor(topic: str) -> dict:
    initial_state = {
        "task": topic,
        "research_text": "",
        "summary": "",
        "routing_log": [],
        "total_tokens": 0,
        "total_seconds": 0.0,
    }
    final_state = graph.invoke(initial_state)
    return {
        "routing_log": final_state["routing_log"],
        "total_tokens": final_state["total_tokens"],
        "total_seconds": final_state["total_seconds"],
    }

if __name__ == "__main__":
    topics = ["climate change", "inflation", "sleep quality"]
    for topic in topics:
        sequential_report = run_sequential(topic)    # from sequential_measure
        supervisor_report = run_supervisor(topic)    # this exercise

        print(f"\n{topic}")
        print(f"  sequential: {sequential_report['total_tokens']} tokens, {sequen
        print(f"  supervisor: {supervisor_report['total_tokens']} tokens, {superv
```

Expected output (exact numbers vary by run):

```
climate change
  sequential: 306 tokens, 2.75s
  supervisor: 306 tokens, 2.81s, routed: ['research', 'write']

inflation
  sequential: 286 tokens, 2.60s
  supervisor: 286 tokens, 2.66s, routed: ['research', 'write']
```

```
sleep quality
  sequential: 325 tokens, 2.90s
  supervisor: 325 tokens, 2.96s, routed: ['research', 'write']
```

Difference from Basic: `run_supervisor()` returns the exact same 3 keys (`total_tokens` , `total_seconds` , plus `routing_log` in place of nothing) as `sequential_measure`'s `run_sequential()` , so the two can be printed side by side without translating field names. Tokens match exactly between sequential and supervisor here — expected, since this `if` -based router adds real wall-clock overhead (the extra node hops) but zero extra tokens, because it never calls the model to decide. That's the exact gap Advanced closes.

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Advanced Version

Approach 1 — an LLM-based supervisor, so the routing decision's real cost shows up

```
import time

def supervisor_llm(state):
    start = time.perf_counter()
    prompt = (
        f"Task: {state['task']}\n"
        f"Research done: {bool(state.get('research_text'))}\n"
        f"Write done: {bool(state.get('summary'))}\n"
        "Reply with exactly one word: RESEARCH, WRITE, or DONE."
    )
    response = model.invoke(prompt)
    elapsed = time.perf_counter() - start
    decision = response.content.strip().upper()

    updates = {
        "total_tokens": state["total_tokens"] + response.usage_metadata["total_to
        "total_seconds": state["total_seconds"] + elapsed,
    }

    if decision == "RESEARCH":
        updates["routing_log"] = state["routing_log"] + ["research"]
        return Command(update=updates, goto="research_node")
    if decision == "WRITE":
        updates["routing_log"] = state["routing_log"] + ["write"]
        return Command(update=updates, goto="write_node")
    return Command(update=updates, goto=END)

builder_llm = StateGraph(SupervisorState)
builder_llm.add_node("supervisor", supervisor_llm)
builder_llm.add_node("research_node", research_node)
builder_llm.add_node("write_node", write_node)
builder_llm.add_edge(START, "supervisor")
graph_llm = builder_llm.compile()
```

```
def run_supervisor_llm(topic: str) -> dict:
    initial_state = {
        "task": topic, "research_text": "", "summary": "",
        "routing_log": [], "total_tokens": 0, "total_seconds": 0.0,
    }
    final_state = graph_llm.invoke(initial_state)
    return {
        "routing_log": final_state["routing_log"],
        "total_tokens": final_state["total_tokens"],
        "total_seconds": final_state["total_seconds"],
    }
```

```
if __name__ == "__main__":
    topics = ["climate change", "inflation", "sleep quality"]
    for topic in topics:
        seq = run_sequential(topic)
        sup_if = run_supervisor(topic)
        sup_llm = run_supervisor_llm(topic)
        print(f"\n{topic}")
        print(f"  sequential:      {seq['total_tokens']}t {seq['total_seconds']}s")
        print(f"  supervisor (if):    {sup_if['total_tokens']}t {sup_if['total_seconds']}s")
        print(f"  supervisor (llm):   {sup_llm['total_tokens']}t {sup_llm['total_seconds']}s")
```

Expected output (exact numbers vary by run):

```
climate change
sequential:      306t 2.75s
supervisor (if): 306t 2.81s
supervisor (llm): 341t 3.62s, routed: ['research', 'write']
```

The LLM-based supervisor calls the model 3 times total (route, route, route-to-done) — roughly 35 extra tokens and 0.8 extra seconds here, entirely from routing decisions that produce zero user-visible work. That's the real, honest cost `paper_design`'s prediction was estimating — an `if`-based router hides it completely.

Approach 2 — a written comparison table across all 3 designs

```
def compare_all(topics: list[str]) -> None:
    print(f"{'topic':<16} {'sequential':>18} {'supervisor(if)':>18} {'supervisor(llm)':>18}")
    for topic in topics:
        seq = run_sequential(topic)
        sup_if = run_supervisor(topic)
        sup_llm = run_supervisor_llm(topic)
        print(
```

```

        f"{topic:<16} "
        f"{seq['total_tokens']}t/{seq['total_seconds']:.1f}s".rjust(18) + " "
        f"{sup_if['total_tokens']}t/{sup_if['total_seconds']:.1f}s".rjust(18)
        f"{sup_llm['total_tokens']}t/{sup_llm['total_seconds']:.1f}s".rjust(18)
    )

compare_all(["climate change", "inflation", "sleep quality"])

```

Expected output (exact numbers vary by run):

topic	sequential	supervisor(if)	supervisor(llm)
climate change	306t/2.8s	306t/2.8s	341t/3.6s
inflation	286t/2.6s	286t/2.7s	319t/3.4s
sleep quality	325t/2.9s	325t/3.0s	362t/3.8s

Real result, versus the `paper_design` guess: sequential and the `if`-based supervisor are nearly identical on tokens (routing there was free) and very close on speed (a few extra node hops, no extra model calls). The LLM-based supervisor is the one that actually matches `paper_design`'s Advanced-level prediction — roughly 10-15% more tokens and 25-30% more wall-clock time than sequential, entirely from a routing decision that, for this fixed two-step task, never had a real choice to make. `paper_design`'s verdict holds: for a task whose order never changes, a supervisor buys nothing except its own overhead.

Difference from Intermediate, and between these 2 Advanced approaches: Intermediate's comparison only shows sequential against a free-routing supervisor, which understates the pattern's real cost — a production supervisor calls the model to route, it doesn't use `if` checks. Approach 1 adds that missing, honest routing cost by building a second supervisor variant that actually pays for its decision. Approach 2 doesn't add new measurement — it lines up all 3 designs' numbers in one table across multiple topics, which is what turns "the LLM supervisor seemed to cost more" into a specific, repeatable percentage you could put in front of the tough senior engineer this document's "How to Practice" step asks you to defend against.

Which one should you actually write? For any real routing decision that could genuinely go more than one way, Advanced Approach 1's LLM-based supervisor is the one whose numbers you can trust — an `if`-based router only tells the truth about tasks so simple they didn't need a supervisor in the first place. For a task this fixed, though, the honest conclusion isn't "use the LLM supervisor" — it's "don't use a supervisor at all here," which is exactly what `paper_design` predicted before any code existed. Keep both supervisor variants in your toolkit: `if`-based routing for genuinely fixed pipelines (call it "sequential with extra steps," not really a supervisor), and LLM-based routing only once the next specialist truly can't be known ahead of time — which is what `ambiguous_routing` builds next.